[CS304] Team Project - Final Delivery (25 points)

> Presentation:
>
> - Week 15 lab session, +0.5 bonus, chance to present at week 16 lecture
> - Week 16 lab session
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> Report submission:
>
> - 06/01/2023 (Thursday) 23:59pm.
> - 40% penalty for late submission.

Team Report (10 points)

In this team report, you should report the following information.

1. Metrics (2 points)

We could use metrics to describe the complexity of your project. Please compute and report the following metrics (you could use whatever tools or libraries for computing the metrics):

- Lines of Code
- Number of modules/subprojects
- Number of packages
- Number of source files
- Number of 3rd party dependencies

We could also use metrics to estimate the maintainability of software systems. Please also answer:

- What is the maintainability of your team project?

To answer this question, first pick one or more proper metrics that could be used to measure software maintainability. Then, you could do one of the following:

- Approach 1: find one or more open source projects that are similar to your team project. Compute the maintainability metrics on the open source projects and on your team project. Compare these metrics, and evaluate the maintainability of your team project based on this comparison.

- Approach 2: if you can't find any open source project that is similar to your team project, you could also provide a detailed analysis on the maintainability of your team project. Specifically, you should use different metrics, statistics, graphs, snapshots, and/or references to back-up your answer.

2. Documentation (2 points)

A well-documented software system is easy to use and maintain. Please demonstrate that your team project has the following two types of documentation:

- Documentation for end users: this could be a `readme` or a `wikipage`, which provides essential information or steps of using the software. End users could use this type of documentation as instructions or user manuals to use your software.

- Documentation for developers: this could be an API documentation that is intended to aid developers. By reading this type of documentation, developers, collaborators, or potential future contributors could understand the design, purpose, and implementation of important code entities (e.g., classes and methods), enabling them to update or extend your software with ease.

It's preferred that you provide URL links to the documentation (e.g., if the documentation is in `.md` or `.html` format and version controlled, you could provide the GitHub link to the resource).

You could also provide snapshots. Yet, if the documentation is too long or scatters around your project, DON'T put all of them in the team report. Instead, you could provide a few snapshots and a textual description describing the documentation.

3. Tests (2 points)

Your project should have **automated** tests for quality assurance. Please describe:

- The technology/tools/frameworks/approaches that you used for automatically testing the project.
- Source code or related artifact for testing.
- Test coverage report.

For the last two items, you could also put snapshots/URL links in the team report.

If, for certain reasons, you only test the project **manually**, please describe the difficulty of test automation, and explain the detailed steps and approaches used in manual testing.

4. Build (2 points)

Your project should have **automated** build process to compile source code, assemble dependencies, and produce executables. Please describe:

- The technology/tools/frameworks/approaches that you used for building the project.
- Tasks executed in a build and the final artifacts produced by a successful build.
- `Buildfile` (e.g., `pom.xml`) or related artifacts/scripts used for building (provide URL links or snapshots).

5. Deloyment (2 points)

A software system needs to be properly deployed to be accessible to the public. Containerization is one of the modern techniques that makes deployment easy.

Please describe:

- The technology/tools/frameworks/approaches that you used for containerizing the project.
- The script or related artifacts (e.g., `Dockerfile`) used for containerization (provide URL links or snapshots).
- A proof of successful containerization (e.g., you could provide a snapshot of a successful docker build).

If, for certain reasons, you cannot containerize the system, please describe the reason why containerization is difficult for your project. You should also provide a detailed description of the deployment approach used for your project.

Project Demonstration (14 points)

In this final presentation, your team should demonstrate the entire running system, focusing on features and usability. For effective demo, your team should **record a video**. Imagining that this video will be played to end users, clients, and potential investors. Your team should demonstrate in the video all notable features of the system, with the purpose of attracting users/investors to use/invest the software system.

The video should conform to the following requirements:

- The video should be no more than 8-minutes long. **You DO NOT need to submit the video to anywhere**. Instead, you only need to play the video during demo. We'll leave 2 minutes for Q&A for each team.
- In the video, you should introduce each feature in a clear, ordered fashion. We require that the team project should support at least 5 distinct, notable features. Hence, you probably want to introduce these features one by one (e.g., the first feature is ...., the second feature is ..., etc.) in the video.
- You DO NOT need to explain implementation details in the video, which is intended to be played to end users/interested parties with no/little technical background.

Points will be deducted if you don't prepare the video or if the video violates the above requirements.

Based on the video demo and the Q&A session, we'll evaluate the following aspects:

- Features (10 points)
- Usability (3 points)
- Presentation & attendance (1 point)

Individual Progress Report (1 point)

Each team member should write and submit a progress report **individually**. In this progress report, you should briefly describe:

- How is the overall team collaboration?
- What is your responsibility for this project?
- What have you contributed to the project?
- What difficulties have you encountered?

Submissions

We will check your team repo for the code and artifacts of your project.

You should also submit a team report and an individual progress report for the final delivery. Both submissions should be made in GitHub Classroom Assignment.

Team report

- Submitted by each team to `GitHub Classroom` -> `Assignments` -> `Team project`.
- Accept this group assignment from this invitation URL
- Join groups (see lab 1 notes).
- Upload the report as `final-report-teamID.md` to the assignment repo. You can find your teamID here.

Individual progress report

- Submitted by everyone to `GitHub Classroom` -> `Assignments` -> `Final-Individual-Report`
- Accept this assignment from this invitation URL.
- Upload the report as `final-report-studentID.md` to the assignment repo.

The file format of report should be `.md` markdown files. Other file format such as `.pdf`, `.docx`, `.txt` will NOT be accepted.